

Task 3 - Set up Crypto++ and OpenSSL

Set up on Windows

Task 3.1: Crypto++ set up

```
tea_simd.cpp    re
tea_simd.o      re
libcryptopp.a   re
libcryptopp.so  re
lsh.h           re
lsh256.cpp      re
lsh256.o        re
lsh256_avx.cpp  re
```

```
passed      verification check against test vector

Testing NaCl library functions...

passed      crypto_box, crypto_box_beforenm, crypto_box_afternm
passed      crypto_box_open, crypto_box_open_afternm
passed      crypto_box_keypair pairwise consistency
passed      crypto_sign, crypto_sign_open, crypto_sign_keypair
passed      crypto_sign_keypair pairwise consistency
```

All tests passed!

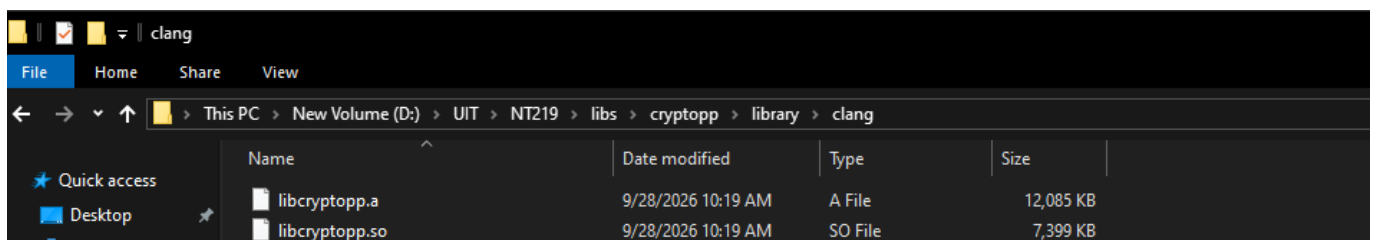
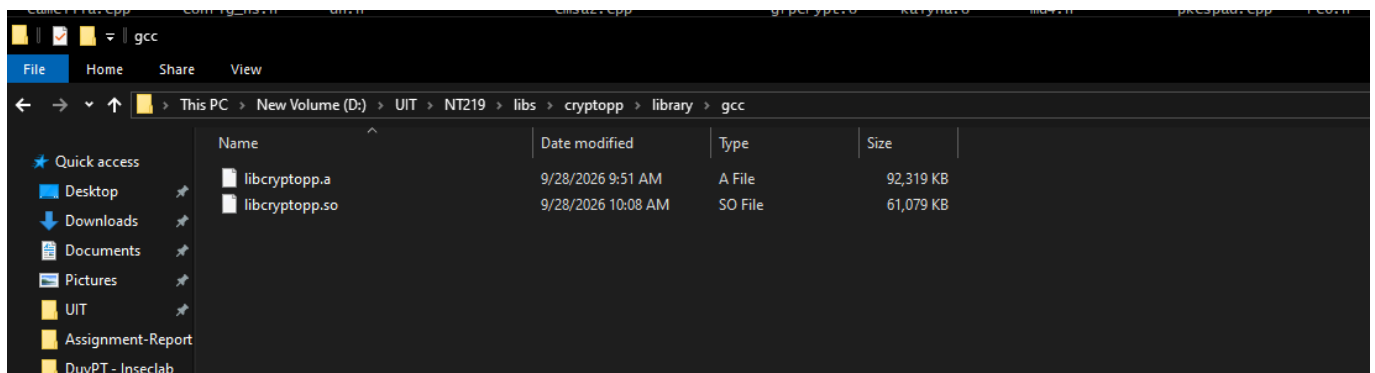
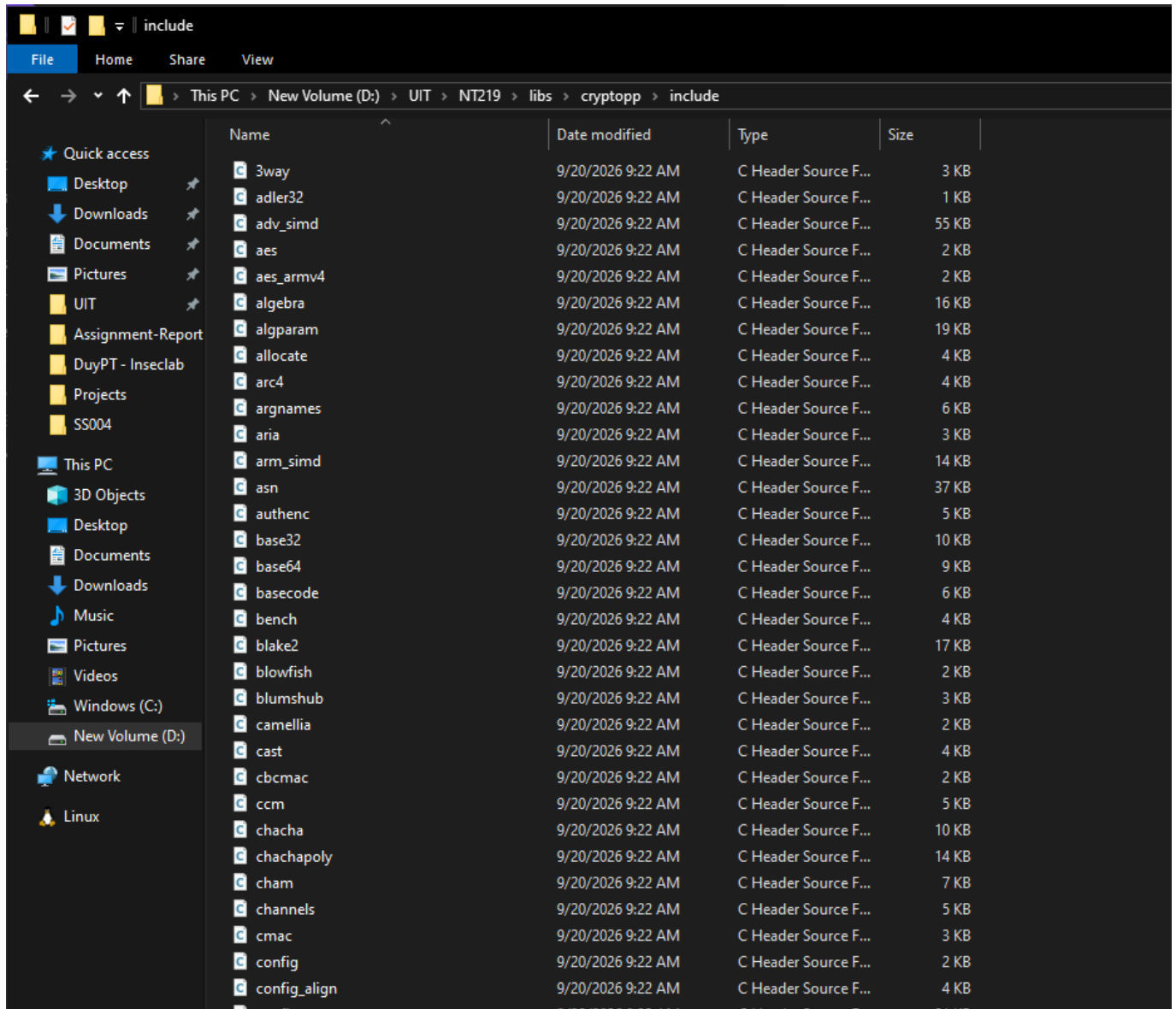
Seed used was 1790563994

Test started at Mon Sep 28 09:53:14 2026

Test ended at Mon Sep 28 09:54:03 2026

DELL@DESKTOP-95JN7H9 MINGW64 /d/UIT/NT219/libs/cryptopp

\$



Output

Show output from: Build

1> speck128_simd.cpp
1> square.cpp
1> squaretb.cpp
1> sse_simd.cpp
1> strciphr.cpp
1> tea.cpp
1> tftables.cpp
1> threefish.cpp
1> tiger.cpp
1> tigertab.cpp
1> ttmac.cpp
1> tweetnacl.cpp
1> twofish.cpp
1> vmac.cpp
1> Generating Code...
1> Compiling...
1> wake.cpp
1> whrpool.cpp
1> xed25519.cpp
1> xtr.cpp
1> xtrcrypt.cpp
1> xts.cpp
1> zdeflate.cpp
1> zinflate.cpp
1> zlib.cpp
1> Generating Code...
1> chacha_avx.cpp
1> dll.cpp
1> iterhash.cpp
1> Generating Code...
1> cryptlib.vcxproj -> D:\UIT\NT219\libs\cryptopp\x64\Output\Release\cryptlib.lib
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
===== Build completed at 10:27 AM and took 01:11.913 minutes =====
|

Output

Show output from: Build

1> oaep.cpp
1> osrng.cpp
1> Generating Code...
1> Compiling...
1> pkcspad.cpp
1> primetab.cpp
1> pssr.cpp
1> pubkey.cpp
1> queue.cpp
1> randpool.cpp
1> rdttables.cpp
1> rijndael.cpp
1> rijndael_simd.cpp
1> rng.cpp
1> rsa.cpp
1> rw.cpp
1> sha.cpp
1> sha_simd.cpp
1> simple.cpp
1> skipjack.cpp
1> sse_simd.cpp
1> strciphr.cpp
1> Generating Code...
1> Creating library D:\UIT\NT219\libs\cryptopp\x64\DLL_Output\Release\cryptopp.lib and object D:\UIT\NT219\libs\cryptopp\x64\DLL_Output\Release\cryptopp.exp
1>LINK : warning LNK4281: undesirable base address 0x42900000 for x64 image; set base address above 4GB for best ASLR optimization
1> cryptdll.vcxproj -> D:\UIT\NT219\libs\cryptopp\x64\DLL_Output\Release\cryptopp.dll
1> Adding MAC to DLL
1> Placing MAC 5F08889C07A5512E3A4EAF83244380662BF3EBC in D:\UIT\NT219\libs\cryptopp\x64\DLL_Output\Release\cryptopp.dll at file offset 1193328 (0x123570).
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
===== Build completed at 10:30 AM and took 02:34.044 minutes =====

Debug

me Share View

> This PC > New Volume (D:) > UIT > NT219 > libs > cryptopp > x64 > Output > Debug

	Name	Date modified	Type	Size
Access	cryptlib.lib	9/28/2026 10:36 AM	Object File Library	162,093 KB
op	cryptlib.pdb	9/28/2026 10:36 AM	Program Debug D...	9,084 KB
loads				
ments				
es				

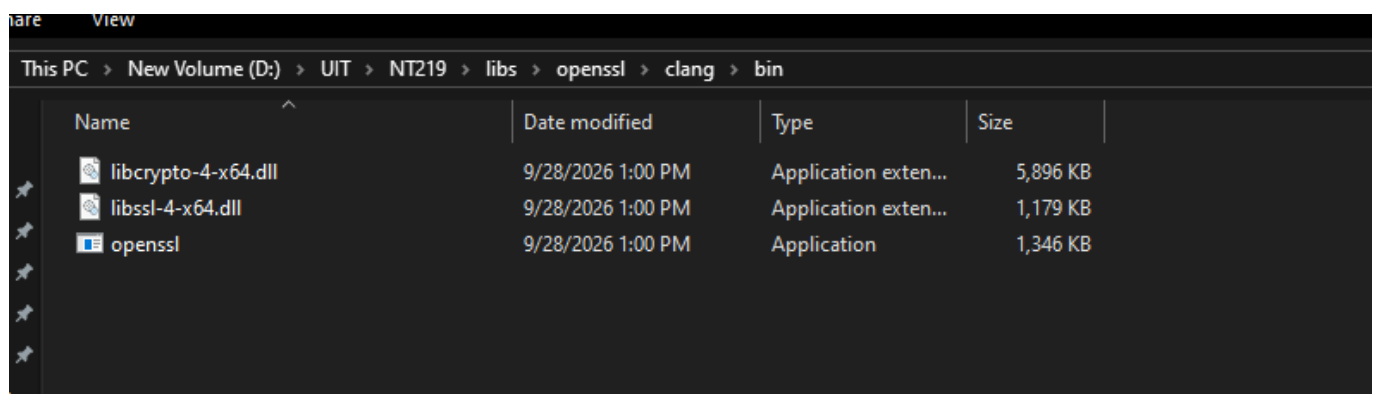
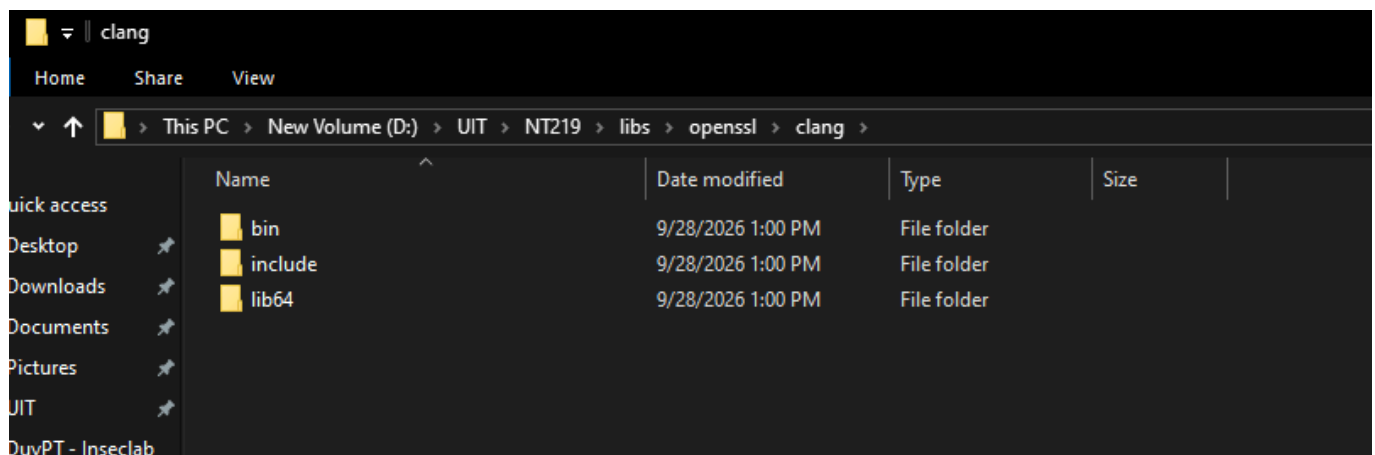
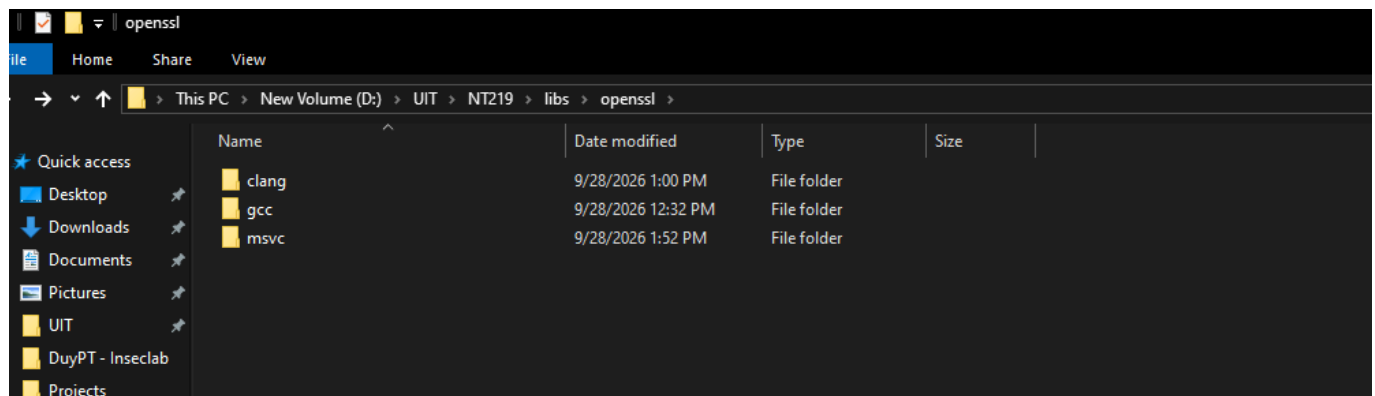
```

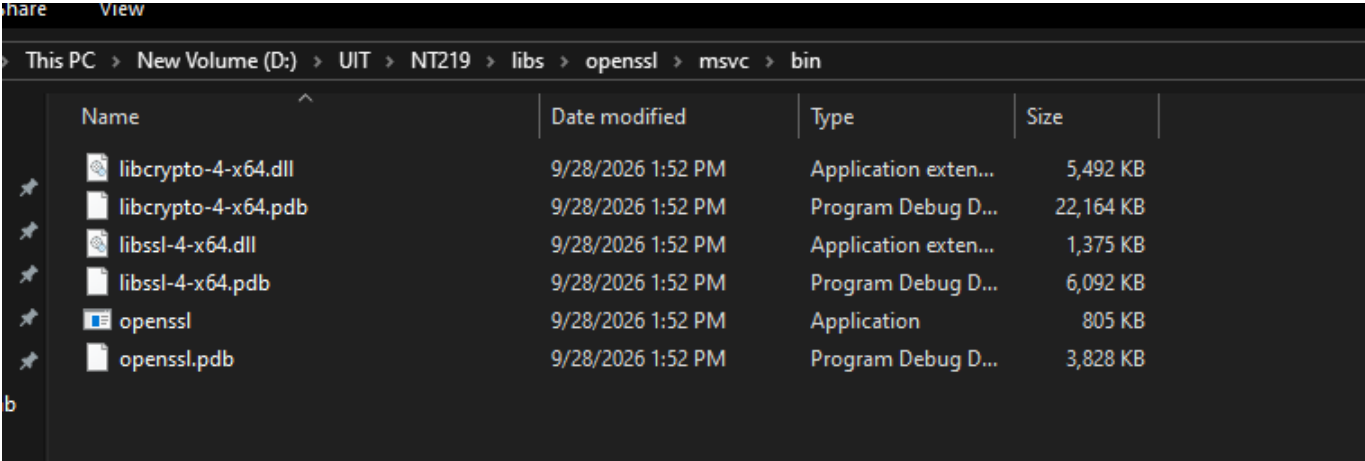
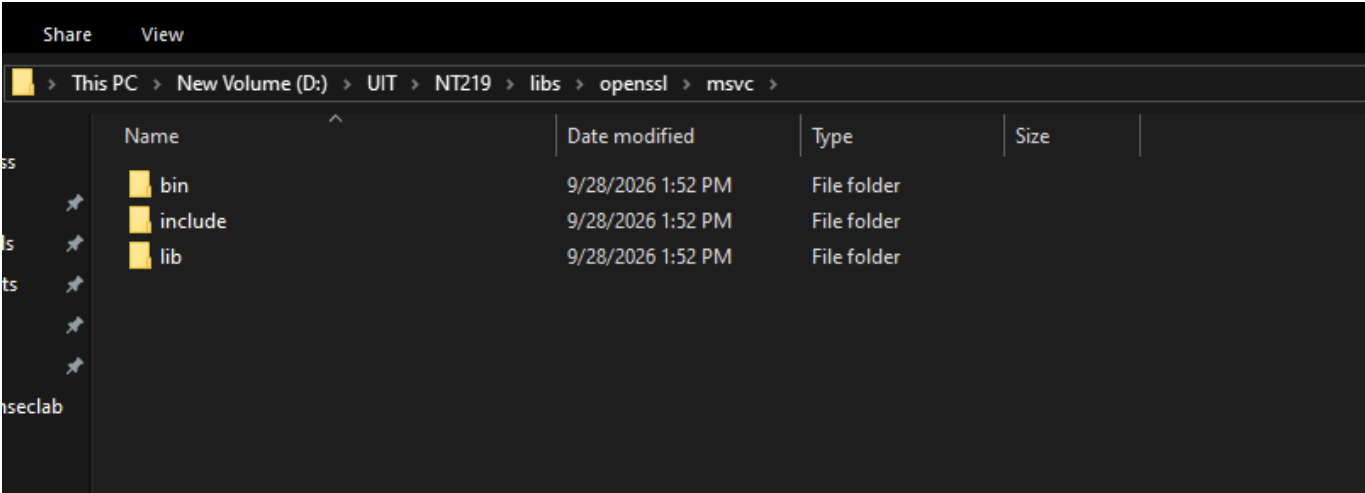
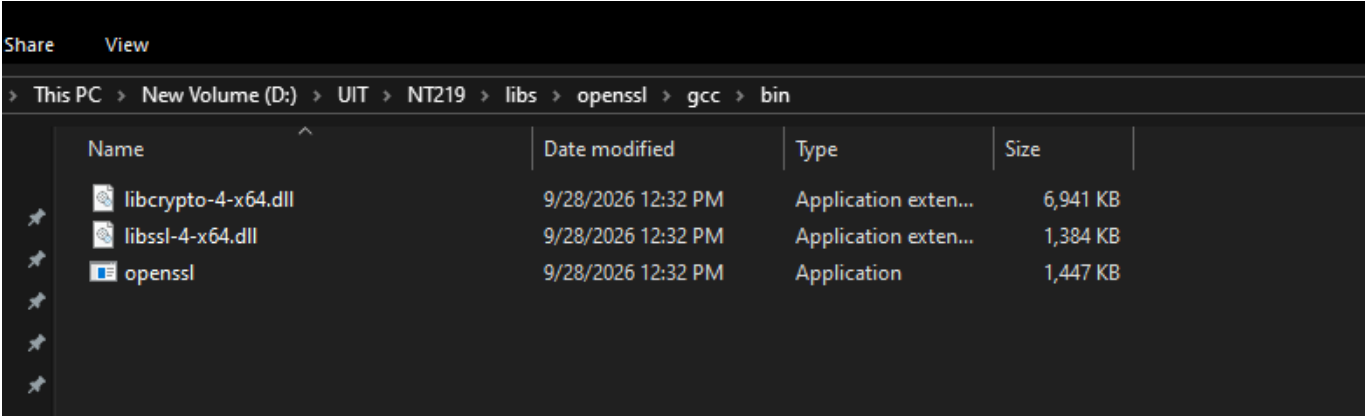
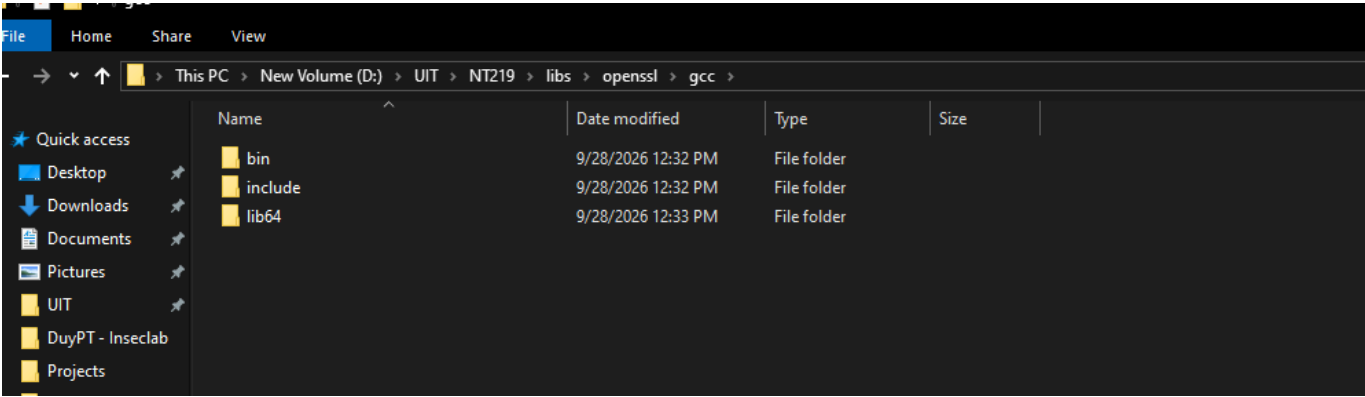
D:\UIT\NT219\libs\cryptopp\library\msvc>ls
cryptlib.lib cryptlib.pdb cryptopp.dll cryptopp.lib cryptopp.pdb

D:\UIT\NT219\libs\cryptopp\library\msvc>ldd cryptopp.dll
ntdll.dll => /c/Windows/SYSTEM32/ntdll.dll (0x7ffe7ebf0000)
KERNEL32.DLL => /c/Windows/System32/KERNEL32.DLL (0x7ffe7ccb0000)
KERNELBASE.dll => /c/Windows/System32/KERNELBASE.dll (0x7ffe7c780000)
msvcrt.dll => /c/Windows/System32/msvcrt.dll (0x7ffe7d3a0000)
ADVAPI32.dll => /c/Windows/System32/ADVAPI32.dll (0x7ffe7d620000)
sechost.dll => /c/Windows/System32/sechost.dll (0x7ffe7da40000)
RPCRT4.dll => /c/Windows/System32/RPCRT4.dll (0x7ffe7d440000)
bcrypt.dll => /c/Windows/System32/bcrypt.dll (0x7ffe7cba0000)

```

Task 3.2: OpenSSL set up





```
YC:\Windows\system32>where openssl
D:\UIT\NT219\libs\openssl\gcc\bin\openssl.exe
D:\UIT\NT219\libs\openssl\clang\bin\openssl.exe
D:\UIT\NT219\libs\openssl\msvc\bin\openssl.exe
```

```
C:\Users\DELL>openssl version
OpenSSL 4.0.2 25 Aug 2026 (Library: OpenSSL 4.0.2 25 Aug 2026)
```

Set up on Ubuntu

Task 3.1: Crypto++ set up

```
dell@dell-Vostro-5481: ~/cryptopp ×

ESIGN validation suite running...

passed    signature key validation
passed    signature and verification
passed    checking invalid signature
passed    verification check against test vector
Generating signature key from seed...
passed    signature key validation
passed    signature and verification
passed    checking invalid signature

x25519 validation suite running...

passed    simple key agreement domain parameters validation
passed    simple key agreement

ed25519 validation suite running...

passed    signature key validation
passed    signature and verification
passed    checking invalid signature
passed    verification check against test vector

Testing NaCl library functions...

passed    crypto_box, crypto_box_beforenm, crypto_box_afternm
passed    crypto_box_open, crypto_box_open_afternm
passed    crypto_box_keypair pairwise consistency
passed    crypto_sign, crypto_sign_open, crypto_sign_keypair
passed    crypto_sign_keypair pairwise consistency

All tests passed!

Seed used was 1790596617
Test started at Mon Sep 28 18:56:57 2026
Test ended at Mon Sep 28 18:57:06 2026
```

```
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc/include/cryptopp$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc/include$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc$ ls
include  lib
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc$ cd lib
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc/lib$ ls
libcryptopp.a  libcryptopp.so
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc/lib$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc$ cd include/cryptopp
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/gcc/include/cryptopp$ ls
3way.h      ccm.h      dh2.h      gcm.h      lubyack.h  pkcsaad.h  seal.h      speck.h
adler32.h   chacha.h   dll.h      gf256.h    luc.h      poly1305.h  secblock.h  square.h
adv_simd.h  chachapoly.h  dmac.h    gf2_32.h   mars.h     polynomi.h  secblockfwd.h  stdcpp.h
aes.h       cham.h     donna.h    gf2n.h     md2.h      ppc_simd.h  seckey.h    strciphr.h
aes_armv4.h channels.h donna_32.h  gfpccrypt.h md4.h      pssr.h      seed.h      tea.h
algebra.h   cmac.h     donna_64.h gost.h     md5.h      pubkey.h    serpent.h   threefish.h
altparam.h  config.h   donna_sse.h gzip.h     mdc.h      pwdbased.h  serpentp.h  tiger.h
allocate.h  config_align.h drbg.h    hashfwd.h  mersenne.h queue.h     sha.h       trap.h
arc4.h      config_asm.h dsa.h     hc128.h    misc.h     rabbit.h    sha1_armv4.h trunhash.h
argnames.h  config_cpu.h eax.h     hc256.h    modarith.h rabin.h     sha256_armv4.h ttmac.h
aria.h      config_cxx.h ec2n.h    hex.h      modes.h    randpool.h  sha3.h      tweetnacl.h
arm_simd.h  config_dll.h eccrypto.h hight.h    modexppc.h rc2.h       sha512_armv4.h twofish.h
asn.h       config_int.h ecp.h     hkdf.h     mqueue.h   rc5.h       shacal2.h   validate.h
authenc.h   config_misc.h ecpoint.h hmac.h     mqv.h      rc6.h       shake.h      vmac.h
base32.h    config_ns.h elgamal.h hmqv.h     naclite.h  rdrand.h    shark.h     wake.h
base64.h    config_os.h emsa2.h   hrtimer.h nbtheory.h resource.h  simeck.h    whrlpool.h
basecode.h  config_ver.h eprecomp.h ida.h      nr.h        rijndael.h  simon.h     words.h
bench.h     cpu.h      esign.h   idea.h     oaep.h     ripemd.h    simple.h    xed25519.h
blake2.h    crc.h      factory.h integer.h  oids.h     rng.h       siphash.h   xtr.h
blowfish.h  cryptlib.h fhmqv.h   iterhash.h osrng.h     rsa.h       skipjack.h  xtrcrypt.h
blumshub.h  darn.h     files.h   kalyna.h   ossig.h    rw.h        sm3.h       xts.h
camellia.h  default.h  filters.h keccak.h   padlkrng.h safer.h      sm4.h       deflate.h
cast.h      des.h      fips140.h lea.h      panama.h   salsa.h     smartptr.h  inflate.h
cbcmac.h    dh.h       fltrimpl.h lsh.h      pch.h      scrypt.h    sosemanuk.h zlib.h
```

```
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang$ cd include/cryptopp
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang/include/cryptopp$ ls
3way.h      ccm.h      dh2.h      gcm.h      lubyack.h  pkcsaad.h  seal.h      speck.h
adler32.h   chacha.h   dll.h      gf256.h    luc.h      poly1305.h  secblock.h  square.h
adv_simd.h  chachapoly.h  dmac.h    gf2_32.h   mars.h     polynomi.h  secblockfwd.h  stdcpp.h
aes.h       cham.h     donna.h    gf2n.h     md2.h      ppc_simd.h  seckey.h    strciphr.h
aes_armv4.h channels.h donna_32.h  gfpccrypt.h md4.h      pssr.h      seed.h      tea.h
algebra.h   cmac.h     donna_64.h gost.h     md5.h      pubkey.h    serpent.h   threefish.h
altparam.h  config.h   donna_sse.h gzip.h     mdc.h      pwdbased.h  serpentp.h  tiger.h
allocate.h  config_align.h drbg.h    hashfwd.h  mersenne.h queue.h     sha.h       trap.h
arc4.h      config_asm.h dsa.h     hc128.h    misc.h     rabbit.h    sha1_armv4.h trunhash.h
argnames.h  config_cpu.h eax.h     hc256.h    modarith.h rabin.h     sha256_armv4.h ttmac.h
aria.h      config_cxx.h ec2n.h    hex.h      modes.h    randpool.h  sha3.h      tweetnacl.h
arm_simd.h  config_dll.h eccrypto.h hight.h    modexppc.h rc2.h       sha512_armv4.h twofish.h
asn.h       config_int.h ecp.h     hkdf.h     mqueue.h   rc5.h       shacal2.h   validate.h
authenc.h   config_misc.h ecpoint.h hmac.h     mqv.h      rc6.h       shake.h      vmac.h
base32.h    config_ns.h elgamal.h hmqv.h     naclite.h  rdrand.h    shark.h     wake.h
base64.h    config_os.h emsa2.h   hrtimer.h nbtheory.h resource.h  simeck.h    whrlpool.h
basecode.h  config_ver.h eprecomp.h ida.h      nr.h        rijndael.h  simon.h     words.h
bench.h     cpu.h      esign.h   idea.h     oaep.h     ripemd.h    simple.h    xed25519.h
blake2.h    crc.h      factory.h integer.h  oids.h     rng.h       siphash.h   xtr.h
blowfish.h  cryptlib.h fhmqv.h   iterhash.h osrng.h     rsa.h       skipjack.h  xtrcrypt.h
blumshub.h  darn.h     files.h   kalyna.h   ossig.h    rw.h        sm3.h       xts.h
camellia.h  default.h  filters.h keccak.h   padlkrng.h safer.h      sm4.h       deflate.h
cast.h      des.h      fips140.h lea.h      panama.h   salsa.h     smartptr.h  inflate.h
cbcmac.h    dh.h       fltrimpl.h lsh.h      pch.h      scrypt.h    sosemanuk.h zlib.h
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang/include/cryptopp$ cd ../.
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang/include$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang$ ls
include  lib
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang$ cd lib
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang/lib$ ls
libcryptopp.a  libcryptopp.so
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/cryptopp/clang/lib$
```

Task 3.2: OpenSSL set up

```

[19:28:24] 80-test_ssl_old.t ..... ok      29971 ms ( 2.84 usr  0.35 sys + 160.80 cusr 52.17 cs
16.16 CPU)
[19:28:24] 90-test_tls13ccs.t ..... ok       816 ms ( 0.07 usr  0.01 sys + 17.65 cusr  6.05 cs
.78 CPU)
[19:28:24] 90-test_tls13secrets.t ..... ok     576 ms ( 0.03 usr  0.01 sys + 17.33 cusr  6.00 cs
.37 CPU)
[19:28:25] 90-test_trace_api.t ..... ok      680 ms ( 0.04 usr  0.01 sys +  1.38 cusr  0.21 cs
.64 CPU)
[19:28:25] 90-test_v3name.t ..... ok     1080 ms ( 0.06 usr  0.01 sys +  1.47 cusr  0.22 cs
.76 CPU)
[19:28:26] 92-test_engine_stubs.t ..... ok      964 ms ( 0.05 usr  0.01 sys +  1.08 cusr  0.16 cs
.30 CPU)
[19:28:26] 95-test_external_cf_quiche.t ..... skipped: No external tests in this configuration
[19:28:26] 95-test_external_ech_nss.t ..... skipped: No external tests in this configuration
[19:28:27] 95-test_external_ech_bssl.t ..... skipped: No external tests in this configuration
[19:28:27] 91-test_pkey_check.t ..... ok     2410 ms ( 0.14 usr  0.03 sys +  3.87 cusr  0.70 cs
.74 CPU)
[19:28:27] 95-test_external_oqsprovider.t ..... skipped: No external tests in this configuration
[19:28:27] 95-test_external_krb5.t ..... skipped: No external tests in this configuration
[19:28:28] 95-test_external_pkcs11_provider.t ..... skipped: No external tests in this configuration
[19:28:28] 95-test_external_rpki-client-portable.t .. skipped: No external tests in this configuration
[19:28:28] 95-test_external_pyca.t ..... skipped: No external tests in this configuration
[19:28:28] 99-test_ecstress.t ..... ok       419 ms ( 0.02 usr  0.00 sys +  0.59 cusr  0.09 cs
.70 CPU)
[19:28:28] 95-test_external_tlzfuzzer.t ..... skipped: No external tests in this configuration
[19:28:35] 80-test_cms.t ..... ok     52413 ms ( 6.03 usr  0.57 sys + 229.63 cusr 66.15 cs
02.38 CPU)
[19:28:38] 90-test_threads.t ..... ok    24340 ms ( 1.43 usr  0.16 sys + 146.10 cusr 17.76 cs
65.45 CPU)
[19:28:42] 90-test_store.t ..... ok     31000 ms ( 2.30 usr  0.21 sys + 163.25 cusr 24.49 cs
90.25 CPU)
[19:28:43] 30-test_evpt.t ..... ok    115534 ms (19.07 usr  1.34 sys + 668.68 cusr 137.24 cs
826.33 CPU)
[19:28:43] 80-test_cmp_http.t ..... ok     64351 ms ( 6.98 usr  0.61 sys + 435.38 cusr 86.50 cs
29.47 CPU)
[19:28:47]
All tests successful.
Files=368, Tests=4388, 227 wallclock secs (25.82 usr  2.35 sys + 1185.51 cusr 233.91 csys = 1447.59 CPU)
Result: PASS
make[2]: Leaving directory '/home/dell/Desktop/openssl_build/openssl-4.0.2'
make[1]: Leaving directory '/home/dell/Desktop/openssl_build/openssl-4.0.2'
/home/dell/Desktop/openssl_build/openssl-4.0.2$ make install

```

```
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc$ ls
bin include lib64
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc$ cd bin
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc/bin$ ls
openssl
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc/bin$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc$ cd include
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc/include$ ls
openssl
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc/include$ cd openssl
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/gcc/include/openssl$ ls
aes.h          cmserr.h       des.h          evp.h          objects.h      rand.h         storeerr.h
asn1.h         comp.h         dh.h          evperr.h       objectsterr.h  randerr.h      symhacks.h
asn1err.h      comperr.h      dherr.h       fips_names.h   ocsp.h         rc2.h          thread.h
asn1t.h        conf.h         dsa.h         fipskey.h      ocspperr.h     rc4.h          tls1.h
async.h        conf_api.h     dsaerr.h      hmac.h         opensslconf.h  rc5.h          trace.h
asynccerr.h    conferr.h      dtls1.h       hpke.h         opensslv.h     ripemd.h       ts.h
bio.h          configuration.h e_os2.h       http.h         ossl_typ.h     rsa.h          tserr.h
bioerr.h       conftypes.h    e_ostime.h    httperr.h      param_build.h  rsaerr.h       txt_db.h
blowfish.h     core.h         ebcddic.h     idea.h         params.h       safestack.h     types.h
bn.h           core_dispatch.h ec.h          indicator.h     pem.h          seed.h         ui.h
bnerr.h        core_names.h   ecdh.h        kdf.h          pem2.h         self_test.h    uierr.h
buffer.h        core_object.h  ecdsa.h       kdferr.h       pemerr.h       sha.h          whrlpool.h
buffererr.h     crmf.h         ecerr.h       lhash.h        pkcs12.h       srp.h          x509.h
byteorder.h     crmferr.h      ech.h         macros.h       pkcs12err.h    srtp.h         x509_acert.h
camellia.h      crypto.h        encoder.h      md2.h          pkcs7.h        ssl.h          x509_vfy.h
cast.h          cryptoerr.h     encodererr.h   md4.h          pkcs7err.h     ssl2.h         x509err.h
cmac.h          cryptoerr_legacy.h engine.h       md5.h          posix_time.h   ssl3.h         x509v3.h
cmp.h           ct.h           engineerr.h    mdc2.h         prov_ssl.h     sslerr.h       x509v3err.h
cmp_util.h      cterr.h        err.h          ml_kem.h       proverr.h      sslerr_legacy.h stack.h
cmperr.h        decoder.h       ess.h          modes.h        provider.h     store.h
cms.h           decodererr.h    esserr.h       obj_mac.h      quic.h
```

```
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang$ ls
bin include lib64
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang$ cd bin
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang/bin$ ls
openssl
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang/bin$ cd ..
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang$ cd include
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang/include$ cd openssl
(base) dell@dell-Vostro-5481:~/Desktop/nt219lib/openssl/clang/include/openssl$ ls
aes.h          cmserr.h       des.h          evp.h          objects.h      rand.h         storeerr.h
asn1.h         comp.h         dh.h          evperr.h       objectsterr.h  randerr.h      symhacks.h
asn1err.h      comperr.h      dherr.h       fips_names.h   ocsp.h         rc2.h          thread.h
asn1t.h        conf.h         dsa.h         fipskey.h      ocspperr.h     rc4.h          tls1.h
async.h        conf_api.h     dsaerr.h      hmac.h         opensslconf.h  rc5.h          trace.h
asynccerr.h    conferr.h      dtls1.h       hpke.h         opensslv.h     ripemd.h       ts.h
bio.h          configuration.h e_os2.h       http.h         ossl_typ.h     rsa.h          tserr.h
bioerr.h       conftypes.h    e_ostime.h    httperr.h      param_build.h  rsaerr.h       txt_db.h
blowfish.h     core.h         ebcddic.h     idea.h         params.h       safestack.h     types.h
bn.h           core_dispatch.h ec.h          indicator.h     pem.h          seed.h         ui.h
bnerr.h        core_names.h   ecdh.h        kdf.h          pem2.h         self_test.h    uierr.h
buffer.h        core_object.h  ecdsa.h       kdferr.h       pemerr.h       sha.h          whrlpool.h
buffererr.h     crmf.h         ecerr.h       lhash.h        pkcs12.h       srp.h          x509.h
byteorder.h     crmferr.h      ech.h         macros.h       pkcs12err.h    srtp.h         x509_acert.h
camellia.h      crypto.h        encoder.h      md2.h          pkcs7.h        ssl.h          x509_vfy.h
cast.h          cryptoerr.h     encodererr.h   md4.h          pkcs7err.h     ssl2.h         x509err.h
cmac.h          cryptoerr_legacy.h engine.h       md5.h          posix_time.h   ssl3.h         x509v3.h
cmp.h           ct.h           engineerr.h    mdc2.h         prov_ssl.h     sslerr.h       x509v3err.h
cmp_util.h      cterr.h        err.h          ml_kem.h       proverr.h      sslerr_legacy.h stack.h
cmperr.h        decoder.h       ess.h          modes.h        provider.h     store.h
cms.h           decodererr.h    esserr.h       obj_mac.h      quic.h
```